

Artificial Intelligence in U.S. Healthcare: Legal framework, liability, and ethical challenges with special focus on pediatric care

AUTHORS:

Torres Ponce, Mariano Enrique

Lawyer (LL.B.), Specialist in Computer Law

ORCID id: [0009-0009-6284-3867](https://orcid.org/0009-0009-6284-3867)

Arana, María Noel

Medical Doctor (MD), Specialist in Pediatrics and Neonatology

ORCID id: [0009-0001-8410-7176](https://orcid.org/0009-0001-8410-7176)

ABSTRACT

This article maps the evolving U.S. regulatory environment for AI in healthcare and translates it into operational guidance with a pediatric lens. We distinguish ML-SaMD from CDS and identify lifecycle control points, including data curation, validation, predetermined change control plans, deployment, and monitoring. We then examine how the 21st Century Cures Act boundary for non-device CDS, FDA device pathways, HIPAA and HITECH privacy rules, and ONC HTI-1 transparency obligations interact in real clinical settings. Building on that analysis, the paper presents a responsibility matrix that allocates duties across manufacturers, institutions, clinicians, and EHR vendors, and proposes a pediatric safety framework that requires validation stratified by age, transparency that is usable in clinical practice, and drift monitoring linked to clear escalation protocols. We argue that existing legal frameworks only partially address algorithmic opacity, fairness, and the long-term outcome risks specific to children, and we offer policy and implementation recommendations to close these gaps. The result is actionable guidance for legal practitioners, health systems, and developers seeking to reconcile safety, equity, and continuous learning in AI-enabled care.

KEYWORDS

Artificial intelligence in healthcare; ML-SaMD; Clinical Decision Support (CDS); Decision Support Interventions (DSI); FDA regulation; ONC HTI-1; HIPAA/HITECH; liability; algorithmic transparency; pediatric safety; fairness.

EXECUTIVE SUMMARY

Background: Artificial intelligence systems have transitioned from experimental tools to deployed clinical instruments across American healthcare, influencing diagnostic decisions, treatment protocols, and resource allocation for millions of patients. This rapid integration has exposed fundamental tensions between innovation imperatives and patient safety obligations, particularly as algorithms trained predominantly on adult populations are applied to pediatric care without adequate validation for developmental differences. While the FDA has established pathways for AI-enabled medical devices and the 21st Century Cures Act has delineated boundaries for clinical decision support, these frameworks were designed for static instruments with predictable performance characteristics and prove inadequate for systems that learn, adapt, and evolve throughout their operational lifetime.

Gap: Current legal and regulatory structures address AI healthcare applications through fragmented oversight spanning FDA device regulation, HIPAA privacy requirements, ONC interoperability standards, and emerging state-level algorithmic accountability initiatives. This fragmentation creates compliance complexity while leaving critical gaps in three domains. First, liability frameworks developed for human decision-making struggle to accommodate scenarios where harm results from algorithmic opacity, training data inadequacies, or bias that emerges only after deployment. Second, transparency requirements conflict with both technical limitations of explainability and competitive pressures to protect proprietary algorithms, creating uncertainty about what constitutes adequate disclosure. Third, pediatric populations face compounded vulnerabilities through data scarcity, developmental heterogeneity, and long-term outcome implications that existing frameworks address only superficially.

Purpose: This article translates the fragmented legal landscape governing healthcare AI into an operational framework that addresses regulatory interpretation, liability allocation, and institutional safeguards with particular attention to pediatric-relevant applications. It examines how the intersection of technical opacity, regulatory evolution, and clinical accountability creates novel challenges for physicians, healthcare institutions, and technology developers. The analysis identifies where existing legal doctrines prove inadequate and proposes concrete implementation strategies to reconcile safety, equity, and continuous learning in AI-enabled care.

Methodology: The analysis synthesizes federal regulatory frameworks including FDA device pathways, HIPAA and HITECH privacy rules, and ONC transparency obligations with traditional medical malpractice and product liability doctrines to map accountability structures. It examines documented cases of algorithmic discrimination and implementation failures including the Optum algorithm, Epic Sepsis Model, IBM Watson for Oncology, and Babylon Health to extract operational lessons. The investigation evaluates how ML-SaMD lifecycle stages from data curation through post-market surveillance create regulatory control points and assesses pediatric-specific technical challenges including data scarcity, developmental variability, and transfer learning limitations. It proposes a responsibility matrix allocating duties across manufacturers, institutions, clinicians, and EHR vendors alongside a pediatric safety framework requiring age-stratified validation and family-centered governance.

Results: The investigation reveals that while significant progress has established basic regulatory pathways, critical gaps persist across multiple domains. Accountability remains contested as traditional liability frameworks encounter limits when harm links to algorithmic bias, training data shortcomings, or inherent system opacity that physicians cannot meaningfully evaluate. The learned intermediary doctrine assumes clinicians can assess medical recommendations, yet AI decision pathways often exceed human interpretive capacity. Pediatric populations face particular vulnerabilities as current approaches treat pediatric applications as adaptations of adult-oriented systems without sufficient validation for developmental differences, data scarcity, or long-term outcome implications. Transparency demands conflict with technical limitations and competitive pressures, creating scenarios where systems may be transparent through documented processes yet inexplicable if logic exceeds human cognitive capacity. Documented cases demonstrate that AI systems can amplify existing healthcare disparities without proactive

design and continuous monitoring, yet regulatory frameworks provide inadequate mechanisms for identifying, measuring, and correcting algorithmic bias particularly for intersectional vulnerabilities in pediatric populations.

Conclusion: Sustainable integration of AI in healthcare demands coordinated evolution across regulatory oversight, institutional governance, and technology development practices. Specialized pediatric AI pathways must recognise developmental considerations and require age-stratified validation beyond simple adult model adaptation. Liability frameworks need reconceptualization to address shared responsibility among physicians, institutions, and technology vendors in AI-assisted care scenarios where traditional negligence doctrines prove inadequate. Transparency requirements must balance meaningful disclosure with technical realities while ensuring patients and providers receive actionable information about AI involvement in care decisions. The window for proactive legal framework development is narrowing as AI adoption consolidates across healthcare settings, making decisive action essential to establish governance structures that promote beneficial innovation while protecting patient safety and rights. The children who will grow up in an AI-enhanced healthcare system deserve frameworks that harness technological capability through legal and ethical structures designed for their unique vulnerabilities and long-term welfare.

TABLE OF CONTENTS

Abstract	1
Keywords	2
Executive Summary	2
A. Introduction	6
B. Technical Foundations for Legal Practitioners	8
B.1. Defining the AI Healthcare Ecosystem	9
B.2. Machine Learning Lifecycle in Healthcare: Regulatory Control Points	10
C. U.S. Regulatory Framework for AI in Healthcare	11
C.1. The 21st Century Cures Act and the CDS Non-Device Exemption	11
C.2. FDA Software as Medical Device Framework Evolution	12
C.3. HIPAA, HITECH, and Pediatric Data Considerations	13
C.4. ONC HTI-1 Decision Support Interventions and EHR Transparency	14
C.5. Institutional Responsibility Matrix	15
D. Liability and Standard of Care	16
D.1. Traditional Medical Malpractice Framework	16
D.2. Product Liability Intersection	18
D.3. Clinical Documentation and Independence Requirements	19
D.4. Emerging Jurisprudential Patterns	20
E. Privacy, Security, and Data Governance	21
E.1. HIPAA Boundaries in AI Training and Deployment	21
E.2. Vendor Risk Management and Clinical SBOMs	22
E.3. Data Minimization versus Learning Optimization	23
E.4. Algorithmic Impact Assessment Frameworks	24
F. Fairness and Non-Discrimination	26
F.1. Documented Cases of Algorithmic Discrimination	26
F.2. Civil Rights and Healthcare Access Obligations	27
F.3. Bias Auditing and Fairness Metrics	28
F.4. Constitutional and Regulatory Constraints	29
F.5. Insurance and Coverage Implications	30
G. AI in Pediatric Care: Special Considerations	30
G.1. Technical Challenges Unique to Pediatric Populations	31
G.2. Legal Framework Specificities for Minors	32
G.3. Current Pediatric AI Applications and Outcomes	33
G.4. Pediatric-Specific Ethical Considerations	34
G.5. Proposed Pediatric AI Safety Framework	35
H. Case Studies and Legal Precedents	37
H.1. Epic Systems Sepsis Algorithm: Lessons from Implementation	37
H.2. IBM Watson for Oncology: Promise versus Reality	38
H.3. Babylon Health Chatbot: Regulatory Response and Liability	39
H.4. Successful Implementations: Radiology CAD Systems	40

I. Recommendations and Implementation Roadmap	41
I.1. For Regulators and Policymakers	42
I.2. For Healthcare Institutions	43
I.3. For Technology Developers	44
I.4. Pediatric-Specific Implementation Priorities	45
J. Conclusions	46
References	48

A. INTRODUCTION

The increasing prominence of artificial intelligence systems in American healthcare generates a fundamental conflict. Tools with opaque internal reasoning are being used to inform high-stakes clinical decisions, including analgesia management in infants. What once appeared as a theoretical concern about black box systems has become a concrete clinical condition. This shift is especially visible in pediatrics, where developmental trajectories and long-term outcomes require a level of precision that error cannot easily tolerate.

The questions raised in this study emerge from a broader transformation of medical responsibility that began with the digitalization of care. During that transition, clinical practice moved into technological environments that reconfigured how responsibility was distributed between professionals and technical systems (Torres Ponce & Arana, 2021). The present analysis extends that line of inquiry into the field of artificial intelligence. The same forces that reshaped presence, trust, and documentation in telemedicine now reach into clinical judgment, affecting liability, transparency, and ethical governance in AI-supported care. These systems do not enter a neutral clinical space. They operate within environments already shaped by information saturation and by preexisting diagnostic

narratives that influence how symptoms are presented and how uncertainty is experienced (Arana, Torres Ponce, & Arroyo Giuliani, 2023).

Since the 21st Century Cures Act of 2016, modest clinical decision support tools have evolved into machine learning systems that approach clinical level performance in several domains (21st Century Cures Act, 2016). Legal and ethical frameworks have struggled to keep pace with this change. Regulatory gaps persist that place both innovation and patient safety at risk. Policy responses have begun to emerge, yet they remain partial and uneven across institutions and jurisdictions.

In pediatrics, these tensions acquire particular weight. Children exhibit developmental variability that challenges models trained largely on adult data, while clinically significant pediatric outcomes are less frequent and large, representative datasets are harder to assemble. Errors in pediatric diagnosis can shape an entire life course. These technical vulnerabilities interact with relational dynamics that already characterize contemporary pediatric care. Caregivers frequently arrive with hypotheses formed in information-saturated environments. AI systems then encounter not only clinical data but also expectations that seek validation (Ramgopal et al., 2023).

This analysis examines three interrelated problems that shape the future of AI in American healthcare. The first concerns the boundary between clinical software and regulated medical devices under FDA oversight and the implications of that boundary for liability and patient protection. The second concerns the content of the standard of care when physicians rely on algorithmic recommendations whose internal logic cannot be fully interpreted by either practitioners or patients. The third concerns the tension between safety, equity, and continuous learning in systems that adapt over time while operating in high-risk pediatric settings.

The regulatory landscape reflects this complexity. Federal oversight through the FDA coexists with data protection regimes under HIPAA and HITECH and with emerging state-level initiatives focused on algorithmic accountability. Traditional theories of medical malpractice strain to address harm linked to algorithmic bias, to limitations in training data, or to the opacity of machine learning models. Insurance practices, informed consent, and quality assurance all face pressure to adapt to a form of care where technical systems participate directly in clinical judgment.

Despite significant progress in establishing regulatory pathways for AI-enabled medical devices, important gaps remain. Pediatric populations continue to receive limited protection from frameworks designed around adult use cases. Standards for meaningful transparency remain uncertain. Liability allocation in cases of algorithmic failure remains fragmented. These problems are compounded by the pace of technological development, which continues to outstrip the capacity of oversight mechanisms to evolve in a coordinated manner.

Regulatory and institutional responses have begun to take shape. Age stratified reporting, alignment between transparency obligations and device labeling, local validation, and pediatric-specific performance metrics all move in the right direction. High risk pediatric contexts also require conservative defaults and sustained human oversight. Taken together, these measures support a form of AI-enabled care that remains anchored in safety and equity rather than in unchecked automation.

The purpose of this article is to translate a fragmented legal landscape into an operational framework for pediatric relevant artificial intelligence. The focus remains on regulatory interpretation, liability allocation, and institutional safeguards capable of supporting safe and accountable use in everyday clinical practice.

B. TECHNICAL FOUNDATIONS FOR LEGAL PRACTITIONERS

To navigate the regulatory, liability, and ethical complexities outlined in the introduction, a clear understanding of the technical underpinnings of healthcare AI is essential, particularly for addressing the distinct requirements of pediatric care. We define the healthcare AI ecosystem, distinguish generative from discriminative systems, outline the ML-SaMD lifecycle from data curation through training and validation to deployment and ongoing monitoring, and clarify the distinction between transparency and explainability. The objective remains descriptive by mapping the concepts that subsequently anchor regulatory classification, liability analysis, and pediatric-specific safeguards.

To facilitate comprehension for legal practitioners, key terms require definition. Machine Learning Software as a Medical Device, or ML-SaMD, refers to AI-driven software performing medical functions. Clinical Decision Support, or CDS, refers to tools that aid

clinician decision-making. Under ONC's HTI-1 certification program, the corresponding certification criterion is Decision Support Interventions, or DSI, which requires transparency about data sources, development and validation methods, and known limitations (Office of the National Coordinator for Health Information Technology, 2023). Predetermined Change Control Plans, or PCCPs, cover pre-approved model updates for ML-SaMD. The Trusted Exchange Framework and Common Agreement, or TEFCA, establishes a nationwide health information exchange framework; it does not itself guarantee security. These concepts anchor the regulatory, liability, and pediatric considerations explored herein.

B.1. DEFINING THE AI HEALTHCARE ECOSYSTEM

Before examining legal frameworks, it is necessary to establish clear definitions that bridge technical precision with legal operability. ML-SaMD refers to machine learning-based medical software that produces diagnostic or therapeutic information without requiring direct physician interpretation for each output (FDA, 2017). This category differs from traditional health IT tools because it functions as a medical device in the regulatory sense. A central regulatory distinction is between discriminative and generative systems. Discriminative models classify inputs within bounded categories, such as radiology applications for tumor detection. Generative models, by contrast, can produce new content, including clinical notes or therapeutic recommendations. This open-ended capacity introduces additional complexity for validation, explainability, and liability allocation.

A further challenge, foreign to traditional medicine, is algorithmic drift. ML-SaMD systems are not static instruments with predictable decay but dynamic entities whose performance can silently erode as patient populations evolve or clinical practices advance. This inherent instability disrupts the traditional regulatory paradigm and requires a framework of continuous monitoring. Real-world performance frequently diverges from controlled trial results, as systems encounter patient populations, clinical workflows, and data quality conditions not represented in development environments.

This opacity challenge has been extensively documented in autonomous systems analysis, where the inability to interpret intermediate neural network layer operations creates fundamental accountability concerns (Torres Ponce, 2019). The phenomenon becomes particularly acute in healthcare settings where clinical decision-making requires not

merely accurate outputs but comprehensible reasoning that physicians can evaluate, validate, and explain to patients. When AI systems process hundreds of variables through multiple hidden layers in ways that defy simple explanation, they challenge the fundamental premise of informed medical judgment, regardless of their accuracy rates. These challenges underscore why lifecycle management, from data curation to post-market monitoring, must be analysed not only as a technical process but as a series of regulatory control points.

B.2. MACHINE LEARNING LIFECYCLE IN HEALTHCARE: REGULATORY CONTROL POINTS

Having established the foundational concepts of the AI healthcare ecosystem, including ML-SaMD definitions and the challenges of explainability, the machine learning operations lifecycle introduces critical regulatory control points, particularly for pediatric applications where data limitations necessitate specialised approaches. Data curation and preprocessing establish the foundation for model accuracy and introduce the first opportunities for bias introduction. Training data selection, cleaning protocols, and feature engineering decisions embed assumptions about patient populations and clinical contexts that persist throughout the system's operational life.

Model training and validation phases require adherence to good machine learning practice, including representative datasets, separation of training and test sets, clinically relevant performance measures, and attention to the interaction between the human user and the model (FDA et al., 2021). For pediatric applications, this stage must also address limited data availability and lower event rates through carefully governed approaches such as transfer learning, synthetic data generation, or federated learning across institutions (Ramgopal et al., 2023).

Model deployment marks the transition from development environment to clinical reality, requiring integration with electronic health record systems, clinical workflows, and existing decision-support tools. The FDA's Predetermined Change Control Plan framework allows for pre-approved modifications during this phase, provided they remain within specified parameters and effectiveness thresholds.

Continuous monitoring and maintenance represent perhaps the most legally complex aspects of ML-SaMD lifecycle management. Continuous monitoring is essential because

AI performance can shift as patient populations or clinical practices evolve. Legal frameworks must account for this dynamism while maintaining safety standards.

This technical complexity illuminates why traditional medical device regulations, designed for static instruments with predictable performance characteristics, may require substantial adaptation for AI systems that learn, adapt, and evolve throughout their operational lifetime.

C. U.S. REGULATORY FRAMEWORK FOR AI IN HEALTHCARE

Building on the technical foundations delineated in the previous section, the regulatory landscape for healthcare AI seeks to balance innovation with patient safety, particularly in pediatric applications where precision is paramount. The regulatory web creates overlapping jurisdictions and compliance obligations that require careful navigation by developers, healthcare institutions, and clinical practitioners seeking to implement AI systems responsibly.

C.1. THE 21ST CENTURY CURES ACT AND THE CDS NON-DEVICE EXEMPTION

Section 3060 of the 21st Century Cures Act excludes a CDS software function from the device definition only when all four statutory criteria are met. The function must not acquire, process, or analyse a medical image or signal from an in vitro diagnostic device or a pattern from a signal acquisition system; it must display, analyse, or print medical information normally communicated between healthcare professionals; it must support or provide recommendations to a healthcare professional; and it must enable that professional to independently review the basis for the recommendation rather than rely primarily on it (21st Century Cures Act, 2016).

However, these seemingly straightforward criteria become complex when applied to modern AI systems. Independent review capability assumes physicians can meaningfully evaluate algorithmic recommendations, yet algorithmic opacity complicates clinical assessment in practice. A radiologist may receive an AI-generated probability score for lung nodule malignancy but may not be able to examine the pixel-level features that contributed to the calculation. Whether the ability to accept or reject the recommendation

constitutes independent review when the underlying logic remains opaque remains an open question.

The rationale requirement proves even more challenging for machine learning systems. Traditional rule-based CDS can provide explicit decision trees showing how patient data leads to recommendations. Machine learning systems complicate this requirement because their reasoning processes often remain opaque, limiting the capacity for meaningful independent review.

Replacement versus supplementation of medical judgment creates a spectrum rather than a binary distinction. While early CDS tools clearly supplemented physician decision-making by flagging drug interactions or suggesting dosing guidelines, modern AI systems can synthesize complex clinical pictures with accuracy exceeding human specialists. Whether physician reliance constitutes replacement or appropriate utilisation of available tools becomes a matter of ongoing regulatory interpretation.

FDA guidance interprets these statutory criteria by focusing on the intended user, the clinical decision involved, the time available for review, and whether the clinician can independently examine the basis of the recommendation (FDA, 2022). Software that fails the statutory criteria may remain a device even when it is described commercially as clinical decision support.

C.2. FDA SOFTWARE AS MEDICAL DEVICE FRAMEWORK EVOLUTION

While the 21st Century Cures Act delineates boundaries for clinical decision support software exempt from device regulation, the FDA's evolving Software as a Medical Device framework provides critical specificity for AI systems, particularly in pediatric applications where risk-based classification ensures patient safety (21 C.F.R. § 860.3, 2024).

Class I devices are lower risk and many are exempt from 510(k) requirements. Basic clinical decision support features such as BMI calculators and simple risk scorecards often receive Class I treatment when regulated.

Class II devices generally require 510(k) clearance demonstrating substantial equivalence to a predicate, and many diagnostic AI systems, including radiological computer-assisted detection and diagnosis tools and some predictive analytics platforms, operate under Class II authorization.

Class III devices require the most rigorous premarket approval and are reserved for the highest-risk indications. Depending on their intended use and risk profile, a narrow subset of AI systems may require Class III oversight, particularly those involving autonomous diagnostic or therapeutic functions used in critical care settings.

The 510(k) pathway can present difficulties for AI-enabled devices when substantial equivalence to a predicate does not adequately capture a novel intended use, technological characteristics, or risk profile. The issue is not that machine-learning methods are inherently incompatible with 510(k), but that meaningful differences may require additional performance evidence or another pathway.

The De Novo pathway, established for novel devices without an appropriate predicate, has become an important route for AI innovation (21 U.S.C. § 360c(f)(2); FDA, 2019). It allows FDA to create new device types with special controls calibrated to risk. Notable De Novo authorizations include IDx-DR for autonomous diabetic retinopathy detection and Caption Guidance for echocardiographic image acquisition.

Predetermined Change Control Plans represent another significant development in oversight of machine-learning-enabled SaMD (FDA, 2024). A PCCP permits predefined, limited algorithm modifications within specified parameters, supported by robust change control, verification and validation, and performance monitoring. This approach recognises the iterative nature of ML while maintaining safety oversight. Effective implementation depends on mature lifecycle processes that may not yet be uniform across institutions.

In December 2024, the FDA issued final guidance on Predetermined Change Control Plans for AI-enabled device software functions. The guidance recommends that a PCCP describe the planned modifications, the methods for developing, validating, and implementing them, and an assessment of their impact. This lifecycle approach is particularly relevant to pediatric applications, where developmental variability and limited datasets make continued validation and post-market monitoring essential (FDA, 2024; Ramgopal et al., 2023).

C.3. HIPAA, HITECH, AND PEDIATRIC DATA CONSIDERATIONS

Beyond the FDA's risk-based classification of AI-enabled medical devices, the integration of these systems into healthcare workflows raises critical data privacy and security

challenges, particularly for pediatric populations subject to stringent protections under HIPAA, HITECH, and state-specific regulations (HIPAA, 1996; HITECH, 2009). Covered entities including hospitals and physician practices must ensure that AI systems maintain appropriate safeguards for protected health information. Business associates such as AI developers and cloud computing providers must execute agreements guaranteeing equivalent protection levels.

Minimum necessary requirements prove particularly challenging for machine learning systems that may require extensive datasets for effective training. Traditional HIPAA guidance, designed for human access to individual patient records, provides limited direction for algorithmic processing of population-level data. The conflict between data minimisation principles and AI effectiveness requirements remains unresolved in current regulatory guidance.

For pediatric populations, HIPAA provides enhanced protections for personal representatives and adolescent healthcare decision-making. Parents typically serve as personal representatives for minor children, yet state law variations create complexity. Some states grant adolescents independent healthcare decision-making authority for reproductive health, mental health, or substance abuse treatment, potentially limiting parental access to AI-generated insights in these domains.

COPPA becomes relevant when an online service directed to children under 13, or an operator with actual knowledge of collection from such children, gathers personal information through patient portals, mobile applications, or educational tools. Healthcare organisations must distinguish COPPA obligations from HIPAA and from state medical-consent rules rather than treating one regime as a substitute for another (COPPA, 1998).

The limitations of HIPAA become even more pronounced in pediatrics, where parental consent, adolescent privacy, and long-term data retention introduce layers of complexity absent in adult populations.

C.4. ONC HTI-1 DECISION SUPPORT INTERVENTIONS AND EHR TRANSPARENCY

The ONC HTI-1 final rule, effective March 11, 2024, replaced the prior clinical decision support certification criterion with the Decision Support Interventions criterion for certified health IT and established baseline transparency requirements for predictive

DSIs. Certified health IT developers must make source attributes available to clinical users so that they can assess fairness, appropriateness, validity, effectiveness, and safety (45 C.F.R. § 170.315(b)(11), 2024; HTI-1 Final Rule, 2024).

Under the DSI criterion, transparency attributes include identification of data sources used in development, the methods for development and validation, known limitations and risks, and performance context. For AI systems, this creates tension between intellectual property protection and regulatory compliance, and developers must balance competitive interests with transparency obligations.

The DSI criterion is directed to certified health IT and the information made available to its users. It does not itself create a general patient right to an explanation of every algorithmic output. Patient-facing disclosure therefore depends on other applicable law, institutional policy, and the clinical context. In pediatric settings, access must also account for parental representation and state-law protections for adolescent confidentiality.

C.5. INSTITUTIONAL RESPONSIBILITY MATRIX

The regulatory framework creates a complex web of overlapping responsibilities among healthcare stakeholders. Manufacturers bear primary responsibility for device safety, effectiveness, and regulatory compliance, including adverse event reporting, quality system maintenance, and post-market surveillance. For AI systems, this extends to algorithm validation, bias mitigation, and performance monitoring across diverse patient populations.

Healthcare institutions serve as device users responsible for appropriate implementation, staff training, and clinical governance. Hospitals must establish AI oversight committees, develop clinical protocols for AI utilisation, and maintain documentation standards for algorithmic decision-making. The lack of standardized institutional frameworks creates significant compliance variation across healthcare systems.

Individual clinicians retain ultimate responsibility for patient care decisions, regardless of AI assistance level. Professional licensing boards have yet to establish comprehensive standards for AI utilisation, leaving practitioners to navigate ethical and legal obligations with limited guidance. The tension between AI reliance and professional judgment represents one of the most significant unresolved issues in current regulatory frameworks.

EHR vendors occupy a unique position as both business associates under HIPAA and potential device manufacturers if their systems include regulated AI functionality. Many EHR companies have integrated third-party AI tools while attempting to avoid direct regulation through careful feature positioning and liability allocation arrangements.

This intricate responsibility distribution creates potential gaps in oversight and accountability, particularly when AI systems span multiple vendors, institutions, and clinical domains. The pediatric care context amplifies these challenges by introducing additional stakeholders, including parents, schools, and specialised care providers, each with distinct legal relationships and information access rights.

D. LIABILITY AND STANDARD OF CARE

While the regulatory framework outlined above establishes critical boundaries for AI deployment in healthcare, questions of liability emerge when these systems fail to meet clinical expectations, particularly in pediatric contexts where accountability is paramount. Medical malpractice frameworks developed for human decision-making encounter unprecedented challenges when applied to algorithmic recommendations, while product liability doctrines struggle to accommodate systems that learn and evolve after deployment. The interplay among physician negligence, institutional oversight responsibilities, and manufacturer duties creates novel liability scenarios that existing legal frameworks address only partially.

Establishing operational standards of care for AI utilisation requires addressing appropriate clinical indications, model validation and drift monitoring procedures, human oversight requirements, comprehensive documentation protocols, and escalation mechanisms for system failures. Pediatric contexts introduce additional complexity through distinct risk profiles, specialised consent frameworks, and long-term outcome considerations that may not manifest until years after AI-assisted care decisions.

D.1. TRADITIONAL MEDICAL MALPRACTICE FRAMEWORK

The foundation of medical liability in the United States rests on negligence principles that require proof of duty, breach of the standard of care, causation, and damages (Restatement (Second) of Torts § 282, 1965). This framework was developed for human decision-

making. Its assumptions become unstable when applied to clinical environments in which algorithmic systems participate directly in diagnostic and therapeutic reasoning.

The physician-patient relationship remains the primary source of legal and ethical duty. The introduction of autonomous or semi-autonomous AI systems complicates this structure. Algorithmic outputs increasingly guide decisions that were once based on human interpretation alone. This creates uncertainty about whether a duty of care can ever arise from the system itself or whether responsibility must always be traced back to human and institutional actors who select, deploy, and rely on these tools.

This redistribution of accountability continues a process that began with the digitalization of medicine during earlier phases of technological transition (Torres Ponce & Arana, 2021). Responsibility shifted away from the isolated clinician toward networks that included platforms, institutions, and regulators. AI-assisted care deepens that shift. Technical systems no longer function only as channels of communication. They shape clinical judgment by filtering data, ranking risks, and generating recommendations that carry practical authority in everyday care.

These tensions are particularly acute in pediatrics. Physicians operate within relationships that include parental authority, evolving minor capacity, and heightened vulnerability. When algorithmic recommendations enter that space, deviations from standard practice acquire a higher liability profile. Decisions are no longer assessed only against professional judgment but also against the perceived objectivity of a technical system that may be treated as an external benchmark.

Standard of care analysis asks what a reasonably prudent physician would do under similar circumstances. As some AI systems demonstrate levels of accuracy that exceed those of human clinicians in narrow tasks, this benchmark begins to shift. Clinical judgment remains central, yet its contours are altered by the presence of tools that can no longer be dismissed as mere aids. The issue is not whether such systems should be used but how their use reshapes professional expectations.

The decline of locality-based standards has reinforced this effect. National benchmarks now reflect common training and broad access to information. AI systems push this convergence further by delivering uniform recommendations across regions. That consistency creates pressure on institutions that lack comparable tools. Differences in technological access begin to translate into differences in perceived quality of care.

Doctrines such as *res ipsa loquitur* also face strain. Negligence may be inferred when harm arises from events that would not ordinarily occur without fault. Algorithmic failures often stem from data bias, model drift, or interactions within complex technical environments. These mechanisms lie beyond the direct control of individual clinicians. Exclusive control becomes harder to establish. Responsibility diffuses across human and technical actors in ways that traditional malpractice law was not designed to capture.

In practice, these dynamics converge in clinical settings already shaped by information saturation and by expectations that seek confirmation rather than exploration (Arana, Torres Ponce, & Arroyo Giuliani, 2023). Physicians do not evaluate algorithmic outputs in a vacuum. They do so under pressure from narratives that predate the consultation and that amplify both perceived risk and legal exposure. AI systems then operate as instruments of formal validation. This further complicates the attribution of fault when outcomes fall short of expectation.

D.2. PRODUCT LIABILITY INTERSECTION

AI medical devices create hybrid liability scenarios that blend medical malpractice with traditional product liability doctrines. While strict liability has long applied to manufacturing defects, design defects, and failures to warn, its application in the AI context presents unique challenges. For instance, issues like corrupted training data, coding errors, or deployment mistakes can be analogized to traditional manufacturing defects. However, the more difficult question arises when machine learning systems function exactly as designed yet produce harmful results that stretch beyond the categories envisioned by classic product liability law.

Design defect analysis typically relies on either the consumer expectation test or the risk-utility balancing test. Neither translates neatly to AI. Ordinary users cannot reasonably form expectations about algorithmic behaviour, rendering the first test ill-suited. Risk-utility balancing, meanwhile, demands a weighing of benefits and risks that is complicated by the dynamic nature of AI, where risks may only surface over time or may affect different patient populations in divergent ways.

The learned intermediary doctrine traditionally shields drug manufacturers from direct liability by placing the duty to warn on physicians rather than patients. AI unsettles this principle. Physicians may lack the technical expertise to meaningfully interpret

algorithmic risks, weakening their ability to serve as effective intermediaries. When an AI system generates recommendations that even trained professionals cannot fully assess, the rationale for the doctrine begins to erode.

Section 230 of the Communications Decency Act does not provide a general liability rule for healthcare AI. Whether it applies depends on the nature of the service, the source of the challenged information, and the theory of liability. Clinical negligence, product defect, and failure-to-warn claims therefore require their own analysis rather than an assumption of platform immunity.

D.3. CLINICAL DOCUMENTATION AND INDEPENDENCE REQUIREMENTS

Proper documentation becomes critical for liability protection when AI systems participate in clinical decision-making. Independent professional judgment must be demonstrated through records showing that physicians considered AI recommendations alongside other clinical factors, rather than reflexively adopting algorithmic outputs. This requires documentation of alternative diagnoses considered, rationale for accepting or rejecting AI recommendations, and integration of AI insights with clinical examination findings.

AI model versioning and change tracking creates new documentation requirements. When AI systems update algorithms, modify training data, or adjust parameters, healthcare providers must maintain records linking patient care decisions to specific system versions. This proves particularly challenging for cloud-based AI services that may update continuously without explicit user notification.

Safety-case methods developed in safety-critical fields offer a structured way to connect claims about safe operation with evidence, assumptions, hazards, and controls (Leveson & Thomas, 2018). In healthcare AI, such documentation can support clinical governance by recording validation, risk mitigation, monitoring, and escalation arrangements. Its legal significance will depend on the quality of the evidence and the duties applicable to the institution, not on the label “safety case” alone.

Audit trail maintenance requires comprehensive logging of AI system inputs, outputs, and decision points. However, privacy requirements and storage costs create practical limitations on audit trail completeness. Legal standards for audit trail sufficiency remain

undefined, creating uncertainty about documentation requirements necessary for liability protection.

In practice, the standard of care requires documenting the intended use of the system, the validated model version, pediatric subgroup performance, and the rationale for acceptance or override of AI recommendations. Drift alerts must be linked to escalation protocols, and patient disclosure should make clear that AI contributed to the clinical decision.

D.4. EMERGING JURISPRUDENTIAL PATTERNS

Direct U.S. case law on injury caused by clinical AI remained limited at the time of this analysis. Adjacent decisions nevertheless illustrate questions likely to arise. *State v. Loomis* considered the use of a proprietary risk-assessment system in sentencing and required warnings about its limitations while allowing the judge to consider the score as one factor among others (*Loomis v. Wisconsin*, 2016). The analogy is imperfect, but it shows how courts may permit algorithmic assistance when human decision-makers retain responsibility and known limitations are disclosed.

Existing doctrine suggests a functional distinction rather than a settled rule. Recommendation-oriented tools are likely to be examined through professional negligence, institutional governance, and warning duties, while more autonomous device functions may bring product design, labeling, and post-market obligations into greater prominence. The allocation will depend on intended use, actual workflow, contractual arrangements, and the causal mechanism of harm.

Carpenter v. United States concerned long-term location records, not health data or medical AI. Its recognition that extensive digital records can reveal deeply private information nevertheless offers a limited analogy for the sensitivity of longitudinal pediatric data (*Carpenter v. United States*, 2018). Any application to consent or health-data governance must still rest on the statutes and doctrines governing the healthcare context.

Because reported judicial decisions directly addressing clinical AI injury were scarce, claims about emerging jurisprudential patterns should remain cautious. The more developed materials were regulatory guidance, scholarly analysis, contractual practice, and disputes involving adjacent algorithmic systems.

The likely near-term path is incremental application of existing negligence, product-liability, privacy, and administrative-law doctrines, with adjustments for opacity, distributed control, and model change. That approach preserves doctrinal continuity but may leave gaps where no actor possesses sufficient information or control to satisfy traditional liability elements.

E. PRIVACY, SECURITY, AND DATA GOVERNANCE

Beyond the liability considerations inherent in AI-assisted care, the integration of healthcare AI raises critical privacy and data governance challenges, particularly in safeguarding sensitive pediatric information under federal and state regulations, institutional ethics policies, and cross-organisational data exchange frameworks. HIPAA and HITECH requirements intersect with institutional policies and data-sharing arrangements to create complex compliance obligations that must accommodate both AI development needs and strong patient privacy protections throughout the entire model lifecycle. De-identification limitations, data minimisation principles, provenance tracking, role-based access controls, encryption standards, audit logging requirements, and incident response procedures all require adaptation for AI-specific risks, including continuous learning dynamics and model retraining demands. The challenge lies in enabling beneficial innovation while maintaining robust patient safeguards and regulatory compliance across dynamic systems that process vast, heterogeneous datasets in ways that traditional privacy frameworks were not designed to anticipate or address effectively.

E.1. HIPAA BOUNDARIES IN AI TRAINING AND DEPLOYMENT

The application of HIPAA to AI systems reveals fundamental tensions between privacy protection and technological advancement. The ethical implications of data governance in AI systems continue a debate that began with the first wave of digital medicine during the pandemic. As observed in *Digital Medicine in Health Emergencies* (Torres Ponce & Arana, 2021), the legitimacy of remote care depends not only on technical safeguards but on the moral integration of privacy, accountability, and trust into the architecture of digital systems. The same principle applies to AI: data protection cannot be treated as an external compliance task but must be embedded in design, development, and deployment. Privacy,

in this sense, is not merely a legal condition but a constitutive element of patient safety and institutional legitimacy.

Protected Health Information includes individually identifiable health information in any form, creating broad coverage that encompasses virtually all data used in healthcare AI development. However, HIPAA's de-identification safe harbor provisions, designed for traditional healthcare uses, prove inadequate for machine learning applications that can re-identify patients through pattern recognition and data correlation.

Research exemptions under HIPAA allow limited PHI use for research purposes, but AI development occupies an ambiguous space between research and commercial product development. Many AI systems begin as research projects but transition to commercial deployment, creating compliance uncertainty about when research protections end and commercial obligations begin. Institutional Review Board approval requirements may apply to AI development using human subjects data, but IRB expertise in AI evaluation remains limited.

Business Associate Agreements require careful drafting to address AI-specific risks. Traditional BAAs contemplate defined uses and disclosures of PHI, but machine learning systems may identify unexpected patterns or correlations that extend beyond originally anticipated uses. Cloud-based AI services present additional challenges because data may be processed across multiple geographic locations and infrastructure providers, each requiring appropriate contractual protections.

Minimum necessary requirements conflict with AI systems' data requirements. Machine learning algorithms often perform better with larger, more comprehensive datasets, while HIPAA requires limiting PHI access to the minimum necessary for intended purposes. This tension becomes acute in pediatric applications where rare conditions require extensive data sharing to achieve statistical significance for training algorithms.

E.2. VENDOR RISK MANAGEMENT AND CLINICAL SBOMS

Software Bill of Materials requirements, inspired by supply chain security frameworks, are emerging as essential tools for healthcare AI governance (Executive Order 14028, 2021). Clinical SBOMs must identify all software components, including training datasets, algorithm libraries, and infrastructure dependencies used in AI systems. This

visibility enables vulnerability management and compliance verification but creates new administrative burdens for healthcare institutions.

Vendor risk assessment for AI systems requires evaluating technical capabilities alongside traditional business factors. Healthcare organisations must assess algorithm validation methodologies, bias testing procedures, and ongoing monitoring capabilities of AI vendors. Multi-vendor integration risks emerge when AI systems from different developers interact within healthcare ecosystems, potentially creating unexpected behaviours or security vulnerabilities.

Data residency and sovereignty requirements vary by jurisdiction and may conflict with AI system architectures that rely on distributed computing resources. European GDPR requirements, applicable to EU patient data regardless of processing location, create compliance obligations that extend beyond U.S. healthcare institutions to their AI vendors and cloud providers (GDPR, 2016). State privacy laws, such as the California Consumer Privacy Act, add additional layers of complexity for healthcare AI systems serving diverse patient populations.

Third-party risk management extends beyond direct AI vendors to include training data providers, cloud infrastructure companies, and integration partners. Each relationship in the AI supply chain presents potential privacy and security risks that healthcare institutions must identify, assess, and mitigate through appropriate contractual and technical controls.

E.3. DATA MINIMIZATION VERSUS LEARNING OPTIMIZATION

Purpose limitation principles require that data collection and use align with specified, legitimate purposes. AI systems challenge these principles because machine learning can identify patterns and relationships that were not anticipated during initial data collection. A pediatric growth monitoring AI might discover correlations between growth patterns and future health risks that extend beyond the original monitoring purpose, raising questions about appropriate use of these insights.

Storage limitation requirements mandate that personal data be kept only as long as necessary for processing purposes. AI systems complicate this principle because model training may require extended data retention, while the resulting models can make predictions without retaining original training data. However, model retraining, bias

correction, and performance monitoring may require access to historical data, creating tension with deletion requirements.

Data quality and accuracy obligations take on new meaning in AI contexts. Training data quality directly affects algorithm performance and fairness, making data accuracy both a privacy and patient safety issue. Right to rectification under various privacy laws allows individuals to correct inaccurate personal data, but correcting training data in deployed AI models may require expensive retraining processes that organisations resist.

Consent mechanisms can be insufficient for AI applications that process data in ways individuals cannot reasonably anticipate or understand. Dynamic consent frameworks attempt to address this challenge by allowing granular, ongoing consent management, but implementation complexity and user burden limit practical adoption. For pediatric populations, consent complexity increases because parents provide initial authorization while children may later seek to modify or withdraw consent as they mature.

In pediatric contexts, transparency obligations extend not only to patients but also to parents or guardians, raising distinctive questions about how algorithmic reasoning is communicated to non-specialist caregivers.

E.4. ALGORITHMIC IMPACT ASSESSMENT FRAMEWORKS

Privacy Impact Assessments systematically evaluate privacy risks associated with new technologies and processes. In healthcare AI, these assessments require enhanced frameworks to address algorithmic bias, the effects of automated decision-making, and long-term privacy implications. Algorithmic Impact Assessments extend beyond privacy to evaluate fairness, accountability, and transparency concerns specific to AI systems, ensuring that healthcare organisations can identify and mitigate risks introduced by complex algorithms.

Data Protection Impact Assessments, mandated by GDPR for high-risk processing activities, offer a robust framework that healthcare organisations increasingly adopt for AI systems, even in jurisdictions where they are not legally required. These assessments must evaluate potential impacts on vulnerable populations, such as pediatric patients, whose unique needs demand specialised considerations. Effective DPIAs for healthcare AI analyse risks related to data scarcity, developmental variability, and informed consent,

while also assessing the adequacy of mitigation measures to protect patient rights and safety.

Continuous monitoring requirements account for the dynamic nature of AI systems, which may evolve in behaviour as they process new data and update algorithms. Unlike traditional privacy assessments conducted once at deployment, AI systems necessitate ongoing evaluation to detect privacy violations, bias amplification, or performance degradation over time. This continuous oversight poses resource challenges for healthcare institutions, particularly those lacking specialised expertise in AI evaluation, necessitating investment in training and infrastructure to ensure compliance.

Transparency reporting mechanisms enable healthcare organisations to communicate AI use policies, performance metrics, and decision-making processes to patients, regulators, and the public. Effective transparency fosters trust and accountability but requires balancing clinical accountability with technical complexity and competitive interests. Healthcare organisations must ensure that transparency reports provide clear, accessible information while protecting proprietary system details and maintaining compliance with regulatory standards.

A critical aspect of transparency is its distinction from explainability. Transparency involves open access to model architecture, training data, and algorithmic processes, allowing regulators and clinicians to verify system design. Explainability requires providing comprehensible reasons for specific AI decisions, enabling physicians to evaluate recommendations effectively. The black box problem occurs when AI systems operate through decision pathways that defy human comprehension, creating scenarios where a system may be transparent through documented processes yet inexplicable if its logic exceeds human cognitive capacity, or explainable through simplified post-hoc rationalizations that obscure proprietary details. In pediatric contexts, where trust and informed consent are critical, transparency and explainability must deliver clinically meaningful information through age-appropriate communication and parental information rights without compromising patient safety or intellectual property. Algorithmic Impact Assessments should incorporate metrics to evaluate both transparency and explainability, ensuring compliance with regulatory and clinical obligations.

Vulnerability response procedures must address technical vulnerabilities and privacy breaches unique to AI systems. Traditional incident response protocols may be inadequate

for AI-related incidents, such as subtle bias introduction, adversarial attacks, or gradual performance degradation. Healthcare organisations require specialised response capabilities that integrate cybersecurity, privacy protection, and clinical safety expertise to effectively manage AI-specific threats and maintain patient trust and safety.

F. FAIRNESS AND NON-DISCRIMINATION

Fairness and non-discrimination in healthcare AI constitute core quality and compliance imperatives that extend beyond technical performance metrics to encompass fundamental civil rights obligations. Bias sources emerge throughout AI system development and deployment, from training data composition and labeling practices to model drift patterns and implementation contexts that can produce disparate outcomes across clinically meaningful subgroups. Healthcare institutions must establish operational controls to measure, monitor, and mitigate discriminatory performance while maintaining clinical effectiveness and regulatory compliance.

Legal obligations prohibiting healthcare discrimination intersect with technical bias evaluation methods to create complex accountability frameworks that require systematic documentation, threshold establishment, and independent review processes. The challenge intensifies in pediatric contexts where developmental factors, family dynamics, and long-term outcome implications compound traditional demographic vulnerability patterns in ways that existing fairness metrics may inadequately capture.

F.1. DOCUMENTED CASES OF ALGORITHMIC DISCRIMINATION

Healthcare AI systems have shown patterns of bias that reproduce and amplify existing disparities in care. The Optum risk prediction model revealed a form of racial distortion in which cost was used as a proxy for health need, leading to systematic underestimation of care requirements for Black patients even when clinical burden was comparable (Obermeyer et al., 2019). The case demonstrated that apparently neutral models can generate outcomes with discriminatory effect under civil rights law when structural inequalities are embedded in the data they process.

IBM Watson for Oncology faced similar limitations. The system was trained largely on data from a single academic cancer center, which meant that its recommendations

reflected the demographic and socioeconomic profile of that institution's patients (Ross & Swetlitz, 2017). When deployed across diverse settings, the system produced advice that often failed to match local patient characteristics, revealing how narrow training environments translate into systematic clinical distortion.

Consumer symptom checkers illustrate a broader risk of uneven performance and unsafe triage advice. Published evaluations have emphasized limitations in evidence quality, validation, and claims of equivalence to clinicians; they do not support treating a single, general pattern of sex-based error as established across all systems (Fraser et al., 2018).

These cases also reveal a less visible source of distortion. Algorithms are not used in neutral clinical settings. They operate within environments shaped by information saturation, anxiety, and preexisting diagnostic narratives that influence how symptoms are framed and how risk is perceived (Arana, Torres Ponce, & Arroyo Giuliani, 2023). When such narratives align with biased model outputs, the combined effect can accelerate diagnostic escalation and reinforce inequitable patterns of care.

Pediatric AI systems face additional layers of vulnerability. Developmental variability, genetic diversity, and social context intersect with limited training data. Models trained in tertiary care centers may perform poorly in community hospitals where patient populations, resource constraints, and clinical practices differ. Smaller pediatric datasets further magnify these problems, particularly when they overrepresent specialised institutions and fail to reflect the broader realities of child health.

F.2. CIVIL RIGHTS AND HEALTHCARE ACCESS OBLIGATIONS

Section 1557 of the Affordable Care Act prohibits specified forms of discrimination in covered health programs and activities receiving federal financial assistance (42 U.S.C. § 18116). When patient-care decision-support tools are used by a covered entity, their design and use must be evaluated within that nondiscrimination framework; the mere appearance of technical neutrality does not remove the need to examine discriminatory effects.

Americans with Disabilities Act requirements apply to healthcare AI systems that must provide effective communication and equal access for individuals with disabilities (ADA, 1990). Voice-based AI systems must accommodate hearing impairments, while visual AI

interfaces must support vision disabilities. Pediatric AI systems must consider developmental disabilities and age-appropriate communication methods.

Title VI of the Civil Rights Act requires that federally funded healthcare programs provide meaningful access for individuals with limited English proficiency (Title VI, 1964). AI systems serving diverse populations must incorporate multilingual capabilities and cultural competency considerations. Training data must reflect linguistic and cultural diversity to avoid systematic bias against non-English speaking populations.

Section 504 of the Rehabilitation Act prohibits disability discrimination in federal programs and requires reasonable accommodations for individuals with disabilities (Section 504, 1973). Healthcare AI systems must be designed and implemented in ways that do not systematically disadvantage individuals with disabilities, either through direct exclusion or through use of proxies that correlate with disability status.

F.3. BIAS AUDITING AND FAIRNESS METRICS

Statistical parity requires that AI system outputs distribute equally across demographic groups. However, this metric may conflict with medical accuracy goals when underlying health risks vary across populations. For example, certain genetic conditions occur more frequently in specific ethnic groups, making statistical parity potentially harmful if it prevents appropriate risk assessment and intervention.

Equalized odds measures require equal true positive and false negative rates across groups, focusing on predictive accuracy rather than outcome distribution. This metric better aligns with medical goals but may still reflect underlying healthcare access disparities that confound the relationship between risk factors and outcomes.

Demographic parity ensures that positive predictions occur at equal rates across groups, while equality of opportunity focuses on equal true positive rates for individuals who should receive positive predictions. The choice of fairness metrics significantly impacts AI system behaviour and requires clinical context-specific evaluation rather than universal application.

Individual fairness approaches evaluate whether similar individuals receive similar predictions, avoiding group-level metrics that may obscure individual discrimination. However, defining similarity in medical contexts requires careful consideration of

clinically relevant factors versus protected characteristics that should not influence treatment decisions.

Intersectionality analysis examines how multiple identity characteristics interact to create unique bias patterns. Pediatric populations present particular intersectionality challenges because age intersects with other demographic factors in ways that may compound discrimination. Young women of color, for example, may face cumulative bias from gender, race, and age factors in AI-driven care recommendations.

F.4. CONSTITUTIONAL AND REGULATORY CONSTRAINTS

Equal Protection Clause analysis applies strict scrutiny to government actions that discriminate based on race or ethnicity, intermediate scrutiny for gender discrimination, and rational basis review for other classifications. Public healthcare institutions using AI systems must ensure that algorithmic decisions can survive appropriate constitutional scrutiny based on the affected groups and interests involved.

Due Process requirements may apply to AI systems that significantly impact patient care decisions, particularly in public healthcare settings. Procedural due process might require notice of AI system use, opportunity to challenge algorithmic recommendations, and access to meaningful human review of automated decisions affecting patient care.

NIST AI Risk Management Framework provides voluntary guidelines for AI bias assessment and mitigation that many healthcare organisations adopt as industry best practices (NIST, 2023). The framework emphasizes continuous monitoring, stakeholder engagement, and documentation requirements that align with legal compliance needs while promoting responsible AI development and deployment.

The NIST AI Risk Management Framework provides a voluntary structure for governing, mapping, measuring, and managing AI risk (NIST, 2023). It is not a binding healthcare regulation, but institutions may use it to organize bias assessment, monitoring, stakeholder engagement, and documentation alongside sector-specific legal obligations.

New York City Local Law 144 concerns automated employment decision tools, not healthcare. Its requirement for a bias audit and specified notices is relevant only as a comparative example of algorithmic-accountability legislation, not as a rule governing clinical AI (NYC Local Law 144, 2023).

F.5. INSURANCE AND COVERAGE IMPLICATIONS

Actuarial fairness principles in insurance pricing may conflict with anti-discrimination requirements when AI systems identify new risk factors that correlate with protected characteristics. Health insurance regulations generally prohibit discrimination based on protected characteristics but allow risk-based pricing for factors considered actuarially justified.

Genetic Information Nondiscrimination Act prohibits health insurance and employment discrimination based on genetic information (GINA, 2008). AI systems that identify genetic risk factors or use genetic proxies in predictions must comply with GINA requirements, which may limit the use of certain algorithmic insights in insurance coverage decisions.

Pre-existing condition protections under the Affordable Care Act prohibit health insurance discrimination based on health status (ACA § 2704, 2010). However, AI systems that predict future health risks may identify individuals likely to develop expensive conditions, creating tension between algorithmic insights and coverage protection requirements.

Medicaid and Medicare coverage decisions increasingly rely on AI systems for fraud detection, utilisation review, and care management. These systems must comply with civil rights requirements and avoid discriminatory impacts on protected populations. Medicare Advantage risk adjustment uses algorithmic methods that have faced scrutiny for potential bias against certain demographic groups.

The intersection of AI capabilities with insurance regulation creates ongoing challenges as algorithms become more sophisticated at risk prediction while legal frameworks struggle to balance actuarial accuracy with anti-discrimination principles. This tension becomes particularly pronounced in pediatric contexts where long-term risk predictions may create lifelong coverage implications based on childhood health data.

G. AI IN PEDIATRIC CARE: SPECIAL CONSIDERATIONS

Pediatric healthcare presents unique challenges for AI implementation that extend far beyond simple population scaling or parameter adjustment. Developmental trajectories, consent and assent frameworks, family-centered care models, and long-term outcome

implications require fundamentally different approaches to AI system design, validation, and governance. Data strategies must accommodate scarce and rapidly evolving patient populations, while validation protocols require calibration to age-specific physiological and behavioural patterns that change continuously throughout childhood and adolescence.

Legal frameworks developed for adult healthcare prove inadequate for pediatric AI applications, where parental authority intersects with emerging adolescent autonomy, long-term privacy implications extend decades into the future, and consent mechanisms must account for developmental capacity limitations. Human oversight and escalation protocols must integrate family-centered decision-making processes while maintaining clinical effectiveness and safety standards appropriate for vulnerable populations whose medical decisions carry lifelong consequences.

Documentation requirements in pediatric AI contexts must capture not only immediate clinical rationale but also developmental appropriateness assessments, family involvement in decision-making, and longitudinal follow-up considerations that may not emerge until years after initial AI-assisted interventions. The intersection of childhood vulnerability, technological complexity, and family dynamics creates governance challenges that existing regulatory frameworks address only superficially.

G.1. TECHNICAL CHALLENGES UNIQUE TO PEDIATRIC POPULATIONS

Data scarcity and precision demands define the central paradox of pediatric AI development. Children represent a substantial share of the population yet exhibit far greater physiological variability due to growth, developmental change, and maturation. This scarcity becomes most acute in rare pediatric conditions, where patient cohorts may number only in the hundreds. Combined with age dependent variability and limited representation in clinical datasets, this makes conventional machine learning approaches statistically fragile and clinically unreliable without pediatric-specific calibration (American Academy of Pediatrics Committee on Bioethics, 2016). Long before the rise of digital medicine, childhood was already understood as a distinct stage of personhood rather than a diminished form of adulthood, a principle that modern AI design must translate into age-stratified modelling (Korczak, 1920/2007).

Developmental heterogeneity challenges systems built around adult physiological assumptions. A cardiac model trained on adult electrocardiographic patterns may misread

normal pediatric rhythms as pathological. Growth prediction systems must account for genetic background, nutrition, and endocrine dynamics that shift rapidly across childhood. Age stratified approaches attempt to manage this complexity, yet they also fragment already limited datasets and reduce statistical power.

Transfer learning from adult populations offers one path forward. It also carries a high risk of systematic distortion. Patterns that predict disease in adults may have different meanings in children. Biomarkers associated with sepsis in adults can reflect ordinary stress responses in younger patients. False positive signals then become more likely. These errors do not only stem from technical miscalibration. They interact with clinical environments in which caregivers arrive with heightened concern and preexisting diagnostic narratives that seek confirmation (Arana, Torres Ponce, & Arroyo Giuliani, 2023). In that setting, algorithmic alerts are more likely to trigger escalation even when clinical probability remains low.

Synthetic data generation has been proposed as a way to expand pediatric training sets through artificial records that preserve statistical properties while protecting privacy (Chen et al., 2021). In pediatrics, this requires accurate modelling of developmental trajectories and rare conditions. These demands exceed the capabilities of most current generative systems, which remain largely experimental in clinical use.

Federated learning allows multiple pediatric institutions to train shared models without exchanging raw patient data (Li et al., 2020). This approach addresses privacy constraints while aggregating experience across sites. Its effectiveness depends on interoperable data standards, compatible infrastructure, and legal frameworks for collaboration that are still unevenly developed.

G.2. LEGAL FRAMEWORK SPECIFICITIES FOR MINORS

Parental permission and child assent require separate analysis when AI systems contribute to pediatric care. The American Academy of Pediatrics recommends involving children and adolescents in decision-making at a developmentally appropriate level while recognising the legal authority and protective responsibilities of parents or guardians (American Academy of Pediatrics Committee on Bioethics, 2016).

Mature-minor doctrines and minor-consent statutes vary by state and clinical context. Mental-health, reproductive-health, and substance-use services may permit adolescents to

consent independently, which can restrict parental access to related AI-generated information. Systems must therefore represent consent status and confidentiality rules rather than assuming that parental authority is uniform.

Emancipated minor status further complicates consent frameworks when adolescents gain legal independence through marriage, military service, or court order. AI systems must accommodate these legal variations while maintaining appropriate clinical oversight and safety protections that recognise adolescents' continuing developmental needs regardless of legal status.

COPPA may apply when a pediatric AI service is directed to children under 13 or when its operator has actual knowledge that it is collecting personal information from such children. Medical necessity is not a general exemption from COPPA. Developers and healthcare organisations must analyse COPPA, HIPAA, and state consent law separately and document which actors and data flows fall within each regime (COPPA, 1998).

Long-term data implications present unique challenges for pediatric AI applications because childhood data may have lifelong significance. Genetic risk predictions, developmental assessments, and behavioural health evaluations conducted in childhood could influence education, employment, and insurance decisions decades later. Legal frameworks must balance immediate clinical benefits with long-term privacy protection for individuals who cannot fully understand future implications of current data use.

G.3. CURRENT PEDIATRIC AI APPLICATIONS AND OUTCOMES

Rare disease diagnosis represents one of the most successful pediatric AI applications. Face2Gene uses facial recognition algorithms to identify genetic syndromes, demonstrating particular effectiveness for conditions with distinctive facial features. The system has contributed to diagnoses for thousands of children with rare conditions, often identifying syndromes that human clinicians missed or required extensive specialist consultation to recognise (Gurovich et al., 2019).

AI-based sepsis prediction has been studied in pediatric and critical-care settings, but performance, workflow integration, and transportability require separate validation. Kamaleswaran and colleagues examined physiomarkers for severe sepsis in the PICU, while the widely deployed Epic Sepsis Model evaluated by Wong and colleagues involved

adult hospitalizations and should not be presented as neonatal evidence (Kamaleswaran et al., 2018; Wong et al., 2021).

AI-assisted autism assessment may combine caregiver questionnaires, clinician observations, and analysis of behavioural information. The FDA granted De Novo classification to the Cognoa ASD Diagnosis Aid in 2021 as an aid, not a stand-alone substitute for clinical diagnosis. Its authorization illustrates both the promise of structured digital assessment and the need to state the intended-use population and limitations precisely (FDA, 2021, DEN200069).

Growth and development monitoring leverages AI to identify deviation from normal developmental trajectories that might indicate underlying conditions requiring intervention. Pediatric growth AI systems can identify endocrine disorders, nutritional deficiencies, and genetic conditions through growth pattern analysis that surpasses traditional percentile charts in sensitivity and specificity.

Congenital heart disease diagnosis benefits from AI systems that analyse fetal ultrasounds, newborn echocardiograms, and chest X-rays to identify structural abnormalities. These applications prove particularly valuable in resource-limited settings where pediatric cardiology expertise is scarce, enabling earlier recognition and referral for life-saving interventions (Arnaout et al., 2021).

Applications such as Face2Gene and the Cognoa ASD Diagnosis Aid illustrate different forms of pediatric AI, but their evidence and regulatory status should not be generalized into a single claim of success through transfer learning. Each system requires evaluation against its own intended use, target population, validation design, and post-deployment performance (Gurovich et al., 2019; FDA, 2021, DEN200069).

G.4. PEDIATRIC-SPECIFIC ETHICAL CONSIDERATIONS

Ethical tensions in pediatric AI-assisted care arise from the interaction between technical recommendations and family mediated decision-making. The best interests standard requires that choices promote child welfare. Algorithmic outputs may conflict with family preferences shaped by cultural values, religious beliefs, or views about quality of life that no model can adequately represent. Even very young children possess forms of expression that deserve recognition within clinical relationships, reinforcing the ethical

importance of child assent in settings where AI contributes to care decisions (Dolto, 1985/1988).

Long term prediction introduces risks that are specific to childhood. When systems label children as being at risk for future conditions that may never materialize, those labels can alter how families, schools, and clinicians perceive the child. Predictions related to neurodevelopmental or mental health outcomes may influence educational pathways, social expectations, and self-understanding in ways that produce lasting effects independent of clinical accuracy. In information saturated environments, such predictions also tend to acquire a narrative force that amplifies anxiety and drives requests for confirmation and intervention (Arana, Torres Ponce, & Arroyo Giuliani, 2023).

Pediatric AI recommendations extend beyond the individual child into the family system. Genetic or risk-based outputs may implicitly include siblings and parents who have not consented to evaluation yet become involved through shared biological or social ties. The distribution of information within families therefore becomes an ethical issue in itself, particularly when probabilistic findings are treated as settled facts.

Developmental capacity remains central to ethical analysis. Children and adolescents differ in their ability to understand risk, to anticipate long-term consequences, and to express preferences about future outcomes. AI systems that support decisions with enduring implications must be deployed within clinical frameworks that recognise these limitations and protect the evolving agency of the child.

School and social integration add another layer of vulnerability. AI generated diagnoses or predictions can influence educational placement, access to services, and peer relationships. Confidentiality and disclosure practices must balance clinical utility against the risk of stigma, especially when labels are applied to conditions that intersect with behaviour, learning, or emotional development.

G.5. PROPOSED PEDIATRIC AI SAFETY FRAMEWORK

Evidence thresholds for pediatric deployment should require higher validation standards than adult AI systems due to developmental considerations and long-term outcome implications. Pediatric AI safety cases must demonstrate effectiveness across age groups, developmental stages, and diverse pediatric populations while establishing appropriate uncertainty thresholds that trigger human oversight or system limitation.

Age-stratified validation requirements should mandate separate testing for neonates, infants, children, and adolescents rather than treating pediatric populations as homogeneous. Each age group presents distinct physiological characteristics, data patterns, and clinical contexts that require separate validation to ensure safety and effectiveness.

Developmental milestone integration should incorporate normal child development knowledge into AI system design and evaluation. Systems must distinguish between developmental variations and pathological conditions, requiring collaboration between AI developers and pediatric development experts during system design and validation phases.

Family-centered design principles should ensure that AI systems support family involvement in pediatric care while respecting appropriate boundaries for adolescent privacy and autonomy. Interface design, information sharing, and decision support features must accommodate family dynamics while maintaining focus on child welfare as the primary consideration.

Long-term outcome monitoring requirements should extend beyond traditional post-market surveillance to track developmental outcomes, educational impacts, and social consequences of AI-assisted pediatric care decisions. This monitoring must continue across childhood and into adulthood to capture delayed effects that may not appear in conventional short-term safety assessments. Longitudinal registries linking childhood AI exposure to adult health outcomes remain an unmet infrastructure need, particularly for rare conditions.

Specialized oversight requirements should establish pediatric AI review committees with expertise in child development, family dynamics, and pediatric medical ethics. These committees should evaluate AI systems for age-appropriate communication, developmental sensitivity, and family impact considerations that general AI oversight committees may lack expertise to assess adequately.

This specialised framework acknowledges that pediatric AI applications require fundamentally different approaches to validation, oversight, and safety monitoring than adult-focused systems, while providing concrete implementation guidance for healthcare institutions, regulators, and technology developers working in pediatric contexts.

H. CASE STUDIES AND LEGAL PRECEDENTS

To illustrate the practical implications of the pediatric-specific challenges outlined in the preceding section, this section analyses case studies of AI implementations in healthcare, drawing lessons applicable to both general and pediatric contexts. Each case demonstrates where existing controls succeeded or failed, how professional norms evolved in response to AI-related incidents, and what operational lessons emerged for documentation, oversight, change control, and stakeholder communication.

Rather than relitigating specific outcomes, these examples distill actionable insights that healthcare institutions, technology developers, and legal practitioners can apply to reduce implementation risks while improving patient care quality. The contrast between failed deployments and successful implementations reveals patterns that predict AI system sustainability and legal risk exposure across diverse clinical contexts.

Performance variations across institutions, workflow integration challenges, regulatory oversight gaps, and liability allocation disputes provide concrete illustrations of theoretical frameworks examined in preceding sections. Documentation deficits, bias manifestations, and consent process failures offer specific guidance for avoiding similar problems in future AI deployments while establishing defensible clinical and legal practices.

H.1. EPIC SYSTEMS SEPSIS ALGORITHM: LESSONS FROM IMPLEMENTATION

The Epic Sepsis Model deployment across hundreds of healthcare systems provides critical insights into real-world AI implementation challenges and legal implications (Wong et al., 2021). Initially lauded for its ability to predict sepsis hours before clinical recognition, subsequent analyses revealed significant performance variations across institutions, patient populations, and clinical contexts that raise fundamental questions about AI system portability and liability allocation.

External validation has shown that performance reported at development sites does not consistently generalize to other institutions with different populations, workflows, and data infrastructure. This variation challenges the traditional medical device paradigm of

consistent performance characteristics and raises questions about the adequacy of single-site validation for multi-institutional deployment.

An external validation at the University of Michigan, covering 38,455 hospitalizations from 2018 to 2019, found that the Epic Sepsis Model showed poor discrimination and calibration. The area under the ROC curve was about 0.63, it captured only around seven percent of sepsis cases that clinicians missed for timely antibiotics, and it triggered alerts for roughly eighteen percent of all inpatients, a volume consistent with alert burden and fatigue (Wong et al., 2021). These results highlight portability limits and the need for local revalidation and governance before widescale deployment.

High alert volumes increased workload and created a meaningful risk of alert fatigue, which can undermine responsiveness to genuine cases and contributes to the portability and governance concerns documented in external evaluations (Habib et al., 2021).

The deployment of the Epic Sepsis Model also raises questions about liability allocation in AI-assisted clinical decision-making. Variability across institutions, together with alert burden, can shape standard-of-care expectations and responsibility distribution among clinicians, hospitals, and vendors, reinforcing the need for robust validation, transparent documentation, and proactive governance (Cohen et al., 2014).

Documentation challenges revealed gaps in clinical record-keeping practices when AI systems participate in decision-making. Many institutions lacked systematic documentation of AI system recommendations, physician responses, and rationale for accepting or rejecting algorithmic suggestions. This documentation deficit complicates both clinical quality improvement and legal defense when AI-assisted care decisions face scrutiny.

H.2. IBM WATSON FOR ONCOLOGY: PROMISE VERSUS REALITY

The implementation challenges observed in the Epic Sepsis Model find parallels in other high-profile AI deployments, such as IBM Watson for Oncology, which further illustrate the complexities of achieving clinical efficacy and regulatory compliance. Initially heralded for its capacity to deliver expert-level cancer treatment recommendations through comprehensive literature synthesis and data integration, Watson's widespread adoption was curtailed by significant operational shortcomings and subsequent market

withdrawal, offering critical insights into the constraints of AI applications and regulatory oversight (Ross & Swetlitz, 2017).

Watson for Oncology was trained and curated around the practices of Memorial Sloan Kettering Cancer Center, and reports from other settings raised concerns about transferability, local treatment protocols, and the quality of some recommendations (Ross & Swetlitz, 2017). These concerns support the need for external validation without establishing that demographic bias was the sole or proven cause of every mismatch.

Watson for Oncology also illustrates the danger of treating the commercial label “clinical decision support” as equivalent to a settled regulatory exemption. Device status depends on intended use and the statutory criteria, while clinical adoption still requires evidence, local governance, and clear limitations even when a product is not actively regulated as a device.

These operational shortcomings raise significant legal considerations regarding responsibility in AI-assisted care. Reports have documented instances where Watson's suggestions deviated from clinical standards, potentially leading to suboptimal treatment decisions. While no malpractice litigation has been publicly documented, these discrepancies highlight the necessity for rigorous validation, transparent performance metrics, and clear protocols to delineate accountability among clinicians, institutions, and technology vendors, ensuring that AI outputs enhance rather than undermine patient outcomes (Somashekhar et al., 2018).

H.3. BABYLON HEALTH CHATBOT: REGULATORY RESPONSE AND LIABILITY

Similar to the clinical integration challenges faced by IBM Watson for Oncology, consumer-facing AI systems like Babylon Health's chatbot highlight additional complexities in regulatory compliance and accountability, particularly when interacting directly with patients, including pediatric populations. Marketed as a triage mechanism capable of assessing symptoms and guiding care pathways, the chatbot faced scrutiny for variable diagnostic accuracy, prompting concerns about its safety and appropriate classification under medical device regulations (Fraser et al., 2018).

Independent evaluations of digital symptom checkers found substantial variation in diagnostic and triage performance and criticized claims of clinician-level equivalence

(Fraser et al., 2018). Regulatory classification depends on intended purpose and functionality; public scrutiny alone should not be described as a formal finding by a regulator unless a specific decision supports that conclusion.

A patient-facing chatbot used by children may trigger COPPA when its operator is covered by that statute, while HIPAA depends on whether the relevant actor is a covered entity or business associate. The two regimes have different scopes, and neither should be assumed to apply merely because the service concerns health.

Documented concerns about symptom-checker accuracy support careful validation, performance disclosure, escalation to human care, and clear responsibility among developers and deploying institutions (Fraser et al., 2018). The business failure of a company may supply context, but it is not itself evidence that a particular clinical output was negligent.

The principal lesson is narrower and more defensible: consumer-facing AI that influences care-seeking can expose users to harm when validation, escalation, and communication are inadequate. Pediatric use raises additional consent, privacy, and safety concerns and therefore warrants conservative deployment and explicit referral pathways.

H.4. SUCCESSFUL IMPLEMENTATIONS: RADIOLOGY CAD SYSTEMS

Unlike consumer-facing systems such as Babylon, which demonstrated significant shortcomings in validation and regulatory compliance, radiology CAD systems demonstrate the potential for successful AI integration in healthcare, particularly through robust validation and clear regulatory pathways that enhance diagnostic precision in both general and pediatric applications (Gulshan et al., 2016). By identifying subtle patterns in imaging data, these tools improve early detection of conditions such as breast cancer and lung nodules, augmenting human expertise in high-stakes diagnostics (Reardon, 2019; Topol, 2019).

The efficacy of modern CAD tools stems from advanced machine learning techniques trained on diverse imaging datasets, incorporating robust validation to ensure consistent performance across varied healthcare settings (Gulshan et al., 2016). This contrasts with earlier AI applications, addressing portability issues seen in systems like Watson or Babylon's chatbot.

Evolving regulatory frameworks have facilitated the safe adoption of CAD systems, with the FDA approving multiple tools under the 510(k) pathway and developing guidance for AI-enabled device software functions. This regulatory clarity mitigates oversight gaps observed in prior clinical decision support tools, enabling efficient deployment.

Effective integration of CAD tools into radiological practice requires balancing algorithmic outputs with independent judgment to avoid diagnostic errors from overreliance or underutilisation (Topol, 2019). Emerging considerations include potential liability risks when radiologists defer excessively to AI, necessitating clear protocols and training to optimize workflow integration and ensure accountability. Ongoing efforts to address biases in training data and ensure equitable performance across diverse populations remain critical, supported by post-market surveillance to maintain safety and efficacy (Reardon, 2019).

I. RECOMMENDATIONS AND IMPLEMENTATION ROADMAP

The lessons derived from the case studies analysed above underscore the need for actionable strategies to address AI's implementation challenges, particularly for safeguarding pediatric populations in healthcare settings. Implementation success depends on establishing clear milestones, accountability mechanisms, and measurable performance indicators that align governance artifacts with existing clinical workflows and quality improvement cycles. Pediatric-specific adaptations must address unique developmental considerations, family-centered care requirements, and long-term outcome monitoring needs that distinguish child healthcare from adult-oriented AI applications.

Moving from principles to repeatable practices demands systematic attention to auditability, scalability, and clinical integration challenges that determine whether AI implementations enhance or disrupt patient care delivery. Regulatory pathways must accommodate AI system dynamism while maintaining safety standards, healthcare institutions require comprehensive governance structures for lifecycle management, and technology developers must prioritize equity, transparency, and clinical usability from initial design through post-market surveillance.

Coordinated stakeholder action becomes essential given the interdependent nature of AI healthcare ecosystems where regulatory gaps, institutional implementation failures, or technology design shortcomings can undermine system-wide safety and effectiveness regardless of individual component quality.

I.1. FOR REGULATORS AND POLICYMAKERS

Establishing specialised pediatric AI review pathways within FDA represents a critical priority that recognises the unique validation requirements and safety considerations for AI systems serving pediatric populations. These pathways should require age-stratified clinical trials, developmental expertise on review panels, and enhanced post-market surveillance for long-term outcomes that may not appear until years after initial deployment.

Developing AI transparency standards must balance intellectual property protection with clinical accountability needs. Regulations should require disclosure of training data sources, validation methodologies, known limitations, and performance metrics across demographic groups, while allowing protection of proprietary algorithms and trade secrets that do not affect clinical interpretation.

Creating liability safe harbors for healthcare providers who follow established AI system use protocols and maintain appropriate oversight responsibilities becomes essential for promoting responsible adoption. Safe harbor provisions should incentivize responsible AI adoption while ensuring that protection requires adherence to evidence-based implementation practices and ongoing competency maintenance.

Mandating algorithmic bias testing as part of regulatory approval processes requires specific requirements for evaluation across racial, ethnic, gender, age, and socioeconomic groups. Testing protocols should address both individual fairness and group fairness metrics, with particular attention to vulnerable populations that may be underrepresented in training data or affected by historical healthcare disparities. Such assessments must also include transparency standards, reproducibility of results, and external auditing mechanisms to ensure accountability and traceability. Establishing federal AI in healthcare research funding specifically directed toward pediatric applications, bias mitigation, and health equity improvement becomes crucial for advancing the field responsibly. Funding priorities should emphasize collaborative research across

institutions, development of synthetic data generation techniques, creation of diverse training datasets that support equitable AI system development, and continuous evaluation frameworks that monitor performance and fairness throughout clinical deployment.

I.2. FOR HEALTHCARE INSTITUTIONS

Developing institutional AI governance committees with expertise spanning clinical medicine, health informatics, bioethics, health law, and health equity represents a foundational requirement for responsible implementation. These committees should establish institutional policies for AI system evaluation, approval, and monitoring while providing ongoing oversight of AI implementation across clinical departments and service lines. Committee membership should include frontline clinicians who will use AI systems, patient representatives, and data scientists capable of interpreting algorithmic performance metrics.

Implementing comprehensive AI system documentation requirements must track AI system involvement in clinical decisions, physician responses to AI recommendations, and rationale for accepting or rejecting algorithmic suggestions. Documentation systems should integrate with electronic health records while supporting both clinical care coordination and legal compliance needs.

Creating AI-specific quality assurance programs becomes essential for monitoring system performance, detecting bias or drift, and evaluating clinical outcomes associated with AI-assisted care. Quality programs should include regular performance audits, patient outcome tracking, and systematic evaluation of AI system impact on healthcare disparities within the institution's patient population. These programs should also incorporate feedback loops between clinicians and developers to promote continuous learning, early detection of anomalies, and integration of post-deployment evidence into system updates. Establishing patient communication protocols for AI system use ensures informed consent, provides clear explanation of AI involvement in care decisions, and offers opt-out mechanisms where clinically appropriate. Communication protocols should be age-appropriate for pediatric populations, culturally sensitive for diverse patient communities, and supported by accessible educational materials. Developing vendor management frameworks specifically for AI system procurement must emphasize ongoing performance monitoring, liability allocation, and data governance requirements. Vendor

agreements should include provisions for algorithm updates, bias mitigation, and performance deterioration remediation while maintaining appropriate intellectual property protections and ensuring compliance with healthcare data standards and ethical principles.

I.3. FOR TECHNOLOGY DEVELOPERS

Prioritizing diverse and representative training datasets that reflect the full spectrum of patients who will encounter AI systems in clinical settings becomes fundamental to equitable system development. Development processes should systematically address known sources of bias in healthcare data while incorporating multiple institutions, geographic regions, and patient populations in training and validation phases.

Designing explainability features that provide clinically meaningful insights into AI system decision-making without requiring extensive technical expertise from healthcare providers proves essential for clinical adoption. Explainability tools should offer different levels of detail appropriate for different clinical contexts and user sophistication while maintaining accuracy in their representation of system behaviour.

Implementing continuous performance monitoring capabilities that detect system drift, bias introduction, or performance degradation in real-world deployment environments represents a critical technical requirement. Monitoring systems should provide automated alerts for performance changes and support rapid response protocols when safety or effectiveness concerns emerge.

Establishing pediatric-specific development protocols addresses the unique challenges of limited training data, developmental variability, and family-centered care requirements. Development protocols should include pediatric expertise throughout the design process and validation approaches that account for age-related physiological and behavioural differences.

Creating transparent update and change management processes maintains system performance while providing appropriate notice to clinical users and regulatory authorities. Change management should balance the benefits of continuous improvement with the need for stability and predictability in clinical care environments.

I.4. PEDIATRIC-SPECIFIC IMPLEMENTATION PRIORITIES

Developing pediatric AI competency requirements for healthcare providers working with children addresses developmental considerations, family dynamics, and age-appropriate communication about AI involvement in care decisions. Competency frameworks should distinguish between different pediatric subspecialties and clinical contexts while ensuring baseline knowledge across all providers caring for children.

Creating specialised informed consent processes for pediatric AI applications must address parental permission, age-appropriate child assent, and long-term implications of AI-assisted care decisions. Consent processes should be developmentally appropriate and culturally sensitive while providing sufficient information for meaningful decision-making by families. Special attention must be given to adolescents in transitional care who may possess legal capacity for independent healthcare decisions in certain jurisdictions.

Establishing pediatric AI research consortiums facilitates data sharing, collaborative algorithm development, and multi-institutional validation studies across children's hospitals and pediatric practices. Consortiums should address technical, legal, and ethical barriers to pediatric AI research while promoting responsible innovation and knowledge sharing.

Developing pediatric AI safety metrics extends beyond traditional clinical endpoints to include developmental outcomes, educational impacts, family satisfaction, and long-term quality of life measures. Safety frameworks should recognise that pediatric AI impacts may not become apparent until years after initial deployment and require ongoing longitudinal monitoring.

Creating pediatric-specific bias evaluation protocols addresses the unique vulnerability factors affecting children, including the intersection of age with other demographic characteristics and the particular importance of equity in childhood healthcare that affects lifelong development and opportunities.

Implementation timeline recommendations suggest a phased approach beginning with low-risk applications in well-resourced settings with extensive oversight, gradually expanding to higher-risk applications and diverse settings as experience and evidence accumulate. This approach should prioritize pediatric safety while avoiding unnecessary delays in beneficial AI applications that could improve child health outcomes.

The successful implementation of these recommendations requires coordinated effort across regulatory agencies, healthcare institutions, technology companies, and professional organisations, with particular attention to the unique needs and vulnerabilities of pediatric populations who stand to benefit significantly from responsible AI innovation in healthcare.

J. CONCLUSIONS

The integration of artificial intelligence in U.S. healthcare has reached an inflection point in which theoretical possibilities have become clinical realities while legal and regulatory frameworks continue to adapt to the demands of this transformation. The analysis shows that although significant progress has been made through the FDA SaMD pathway and the clinical decision support provisions of the 21st Century Cures Act, important gaps remain that place both innovation and patient safety under strain.

Accountability remains a central challenge. Liability doctrines built around human decision-making encounter structural limits when harm is linked to algorithmic bias, inadequate training data, or the opacity of machine learning systems. The learned intermediary doctrine presumes that clinicians can meaningfully evaluate medical recommendations. In AI-assisted care this assumption becomes fragile when decision pathways are difficult to interpret in practice. Uncertainty then emerges over how responsibility should be allocated among physicians, healthcare institutions, and technology vendors when outcomes deviate from expectations.

Pediatric populations face forms of vulnerability that existing frameworks address only incompletely. Scarce training data, developmental variability, and the weight of long-term outcomes combine to create risk profiles that differ fundamentally from adult care. Many pediatric AI applications continue to rely on models developed from adult populations without sufficient developmental validation. Regulatory approaches that treat pediatric use as a derivative of adult systems fail to capture these distinctions and do not reflect the full scope of clinical and ethical exposure involved.

The transparency paradox defines another core tension. Clinicians and patients require meaningful insight into how algorithmic recommendations are generated in order to sustain professional responsibility and informed consent. At the same time, the technical

architecture of advanced models and the commercial context in which they are developed limit the degree of explainability that can be provided. Legal frameworks must navigate this tension in a way that supports trust and accountability without foreclosing beneficial innovation. This challenge is intensified in clinical environments already shaped by information saturation and by expectations that seek confirmation rather than deliberation, conditions that increase the weight placed on algorithmic outputs in everyday decision-making.

Equity concerns permeate every stage of AI implementation, from the composition of training datasets through deployment practices and outcome assessment. The analysis demonstrates that algorithmic systems can reproduce and amplify existing disparities unless proactive design and continuous monitoring are in place. Current regulatory mechanisms remain insufficient to ensure consistent detection and correction of bias, particularly in pediatric populations where social and developmental factors intersect to produce compounded forms of vulnerability.

Regulatory fragmentation across FDA oversight, HHS civil rights enforcement, state professional licensing, and emerging algorithmic accountability regimes adds further complexity. Smaller institutions and pediatric specialty practices face particular burdens as they attempt to comply with overlapping and sometimes inconsistent requirements. This fragmentation reflects a broader global pattern. The European Union has adopted a horizontal risk-based approach under its AI Act, while Canada has emphasized lifecycle governance through Health Canada. The absence of unified federal guidance in the United States leaves healthcare providers navigating uncertainty as they seek to deploy AI responsibly.

Several priorities emerge for the evolution of legal and regulatory frameworks. Pediatric specific oversight pathways should recognise developmental realities and require age-stratified validation. Liability regimes need to reflect shared responsibility across human and technical actors rather than relying on assumptions inherited from pre-digital medicine. Transparency obligations must provide actionable information to clinicians and families while remaining grounded in technical feasibility.

Progress will depend on coordinated action. Regulators must craft oversight structures capable of addressing system dynamism without eroding safety. Healthcare institutions require governance models that can evaluate and monitor AI across its operational life.

Technology developers must integrate equity, transparency, and pediatric considerations from design through post-market surveillance.

This analysis is limited by the pace of technological and regulatory change and by the small body of jurisprudence directly addressing healthcare AI. International developments may also influence domestic approaches in ways that cannot yet be fully anticipated.

Future research should prioritize longitudinal evaluation of pediatric AI outcomes, the development of practical methods for bias detection and mitigation, and empirical study of alternative liability allocation models. Legal scholarship and technical research will need to remain in dialogue as these systems continue to reshape clinical practice.

The potential benefits of healthcare AI are substantial. Improved diagnostic accuracy, reduced treatment error, and expanded access to specialised expertise offer particular promise for underserved pediatric populations. These gains will not be realized if legal and ethical frameworks fail to keep pace with deployment.

The window for proactive governance is narrowing as AI becomes embedded across healthcare settings. Policymakers, clinicians, and developers share responsibility for ensuring that this transformation advances patient welfare rather than undermining it. The children who will grow up within AI mediated systems of care depend on the choices made at this stage.

The future of healthcare AI will be shaped not only by technical capability but by the quality of the legal and ethical structures that guide its use. The task now is to ensure that this power is exercised with care, accountability, and respect for the human contexts in which it operates.

REFERENCES

21st Century Cures Act, Pub. L. No. 114-255, § 3060 (2016). [Source](#)

42 U.S.C. § 18116. [Source](#)

45 C.F.R. § 170.315(b)(11) (2024). [Source](#)

American Academy of Pediatrics Committee on Bioethics. (2016). Informed consent in decision making in pediatric practice. *Pediatrics*, 138(2), e20161484.

<https://doi.org/10.1542/peds.2016-1484>

Americans with Disabilities Act, 42 U.S.C. §§ 12101–12213 (1990). [Source](#)

Arana, M. N., Torres Ponce, M. E., & Arroyo Giuliani, S. S. (2023, July 20). Overdiagnosis by information overload: Self-diagnosis, health anxiety, and the clinical visit as a validation encounter [Preprint]. SSRN. <https://doi.org/10.2139/ssrn.6056414>

Arnaut, R., Curran, L., Zhao, Y., Levine, J. C., Chinn, E., & Moon-Grady, A. J. (2021). An ensemble of neural networks provides expert-level prenatal detection of complex congenital heart disease. *Nature Medicine*, 27(5), 882–891. <https://doi.org/10.1038/s41591-021-01342-5>

Carpenter v. United States, 585 U.S. 296 (2018). [Source](#)

Chen, R. J., Lu, M. Y., Chen, T. Y., Williamson, D. F. K., & Mahmood, F. (2021). Synthetic data in machine learning for medicine and healthcare. *Nature Biomedical Engineering*, 5(6), 493–497. <https://doi.org/10.1038/s41551-021-00751-8>

Children’s Online Privacy Protection Act, 15 U.S.C. §§ 6501–6506 (1998); Children’s Online Privacy Protection Rule, 16 C.F.R. part 312. [Source](#)

Code of Federal Regulations, 21 C.F.R. § 860.3(c)(1)–(3) (2024). [Source](#)

Cohen, I. G., Amarasingham, R., Shah, A., Xie, B., & Lo, B. (2014). The legal and ethical concerns that arise from using complex predictive analytics in health care. *Health Affairs*, 33(7), 1139–1147. <https://doi.org/10.1377/hlthaff.2014.0048>

Dolto, F. (1988). The case of the child: The child as a subject (S. Fairfield, Trans.). Karnac. (Original work published 1985) [Source](#)

Executive Order No. 14028, 86 Fed. Reg. 26633 (May 12, 2021). [Source](#)

Food and Drug Administration. (2017). Software as a medical device (SaMD): Clinical evaluation. U.S. Department of Health and Human Services. [Source](#)

Food and Drug Administration. (2019). De Novo classification process (evaluation of automatic class III designation). U.S. Department of Health and Human Services. [Source](#)

Food and Drug Administration, Health Canada, & Medicines and Healthcare products Regulatory Agency. (2021). Good machine learning practice for medical device development: Guiding principles. [Source](#)

Food and Drug Administration. (2022). Clinical decision support software: Guidance for industry and Food and Drug Administration staff. U.S. Department of Health and Human Services. [Source](#)

Food and Drug Administration. (2024). Marketing submission recommendations for a predetermined change control plan for artificial intelligence-enabled device software functions. U.S. Department of Health and Human Services. [Source](#)

Food and Drug Administration. (2021). Cognoa ASD Diagnosis Aid (DEN200069), De Novo classification request. [Source](#)

Fraser, H., Coiera, E., & Wong, D. (2018). Safety of patient-facing digital symptom checkers. *The Lancet*, 392(10161), 2263–2264. [https://doi.org/10.1016/S0140-6736\(18\)32819-8](https://doi.org/10.1016/S0140-6736(18)32819-8)

General Data Protection Regulation, Regulation (EU) 2016/679 (2016). [Source](#)

Genetic Information Nondiscrimination Act, 42 U.S.C. §§ 2000ff–2000ff-11 (2008). [Source](#)

Gulshan, V., Peng, L., Coram, M., et al. (2016). Development and validation of a deep learning algorithm for detection of diabetic retinopathy in retinal fundus photographs. *JAMA*, 316(22), 2402–2410. <https://doi.org/10.1001/jama.2016.17216>

Gurovich, Y., Hanani, Y., Bar, O., et al. (2019). Identifying facial phenotypes of genetic disorders using deep learning. *Nature Medicine*, 25(1), 60–64. <https://doi.org/10.1038/s41591-018-0279-0>

Habib, A. R., Lin, A. L., & Grant, R. W. (2021). The Epic Sepsis Model falls short. *JAMA Internal Medicine*, 181(8), 1040–1041. <https://doi.org/10.1001/jamainternmed.2021.3333>

Health Data, Technology, and Interoperability: Certification Program Updates, Algorithm Transparency, and Information Sharing, 89 Fed. Reg. 1196 (Jan. 9, 2024). [Source](#)

Health Information Technology for Economic and Clinical Health Act, Pub. L. No. 111-5 (2009). [Source](#)

Health Insurance Portability and Accountability Act, Pub. L. No. 104-191 (1996). [Source](#)

Kamaleswaran, R., Akbilgic, O., Hallman, M. A., et al. (2018). Applying artificial intelligence to identify physiomarkers predicting severe sepsis in the PICU. *Pediatric Critical Care Medicine*, 19(10), e495–e503. <https://doi.org/10.1097/PCC.0000000000001666>

Korczak, J. (2007). *How to love a child* (E. P. Kulawiec, Trans.). Farrar, Straus and Giroux. (Original work published 1920) [Source](#)

Leveson, N. G., & Thomas, J. P. (2018). STPA handbook. MIT. [Source](#)

Li, T., Sahu, A. K., Talwalkar, A., & Smith, V. (2020). Federated learning. *IEEE Signal Processing Magazine*, 37(3), 50–60. <https://doi.org/10.1109/MSP.2020.2975749>

State v. Loomis, 881 N.W.2d 749 (Wis. 2016). [Source](#)

National Institute of Standards and Technology. (2023). Artificial intelligence risk management framework (AI RMF 1.0). <https://doi.org/10.6028/NIST.AI.100-1>

New York City Local Law 144, N.Y.C. Admin. Code §§ 20-870–20-874 (2021; rules effective 2023). [Source](#)

Obermeyer, Z., Powers, B., Vogeli, C., & Mullainathan, S. (2019). Dissecting racial bias in an algorithm used to manage the health of populations. *Science*, 366(6464), 447–453. <https://doi.org/10.1126/science.aax2342>

Office of the National Coordinator for Health Information Technology. (2024). HTI-1 final rule: Certification program updates, algorithm transparency, and information sharing. U.S. Department of Health and Human Services. [Source](#)

Patient Protection and Affordable Care Act § 2704, 42 U.S.C. § 300gg-3 (2010). [Source](#)

Ramgopal, S., Sanchez-Pinto, L. N., Horvat, C. M., Carroll, M. S., Luo, Y., & Florin, T. A. (2023). Artificial intelligence-based clinical decision support in pediatrics. *Pediatric Research*, 93(2), 334–341. <https://doi.org/10.1038/s41390-022-02226-1>

Reardon, S. (2019). Rise of the robot radiologists. *Nature*, 574(7777), S24–S26. <https://doi.org/10.1038/d41586-019-03847-z>

Restatement (Second) of Torts § 282 (American Law Institute 1965). [Source](#)

Ross, C., & Swetlitz, I. (2017, September 5). IBM pitched Watson as a revolution in cancer care. It's nowhere close. *STAT*. [Source](#)

Section 504 of the Rehabilitation Act of 1973, 29 U.S.C. § 794. [Source](#)

Somashekhar, S. P., Sepúlveda, M. J., Puglielli, S., et al. (2018). Watson for Oncology and breast cancer treatment recommendations. *Annals of Oncology*, 29(2), 418–423. <https://doi.org/10.1093/annonc/mdx781>

Title VI of the Civil Rights Act of 1964, 42 U.S.C. § 2000d. [Source](#)

Topol, E. J. (2019). High-performance medicine: The convergence of human and artificial intelligence. *Nature Medicine*, 25(1), 44–56. <https://doi.org/10.1038/s41591-018-0300-7>

Torres Ponce, M. E. (2019). Derechos y desafíos de la inteligencia artificial. *Ciencia y Técnica Administrativa*, 18(1). [Source](#)

- Torres Ponce, M. E., & Arana, M. N. (2021, November 23). Digital medicine in health emergencies: Redefining clinical and legal responsibility [Preprint]. SSRN.
<https://doi.org/10.2139/ssrn.5695082>
- Wong, A., Otles, E., Donnelly, J. P., et al. (2021). External validation of a widely implemented proprietary sepsis prediction model in hospitalized patients. *JAMA Internal Medicine*, 181(8), 1065–1070.
<https://doi.org/10.1001/jamainternmed.2021.2626>